



RAFFLES INSTITUTION
2019 YEAR 6 PRELIMINARY EXAMINATION
Higher 3

MATHEMATICS

9820/01

26 September 2019

3 hours

Additional materials: Answer Booklet
 List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, submit all answer booklets used.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **7** printed pages.

RAFFLES INSTITUTION
Mathematics Department

1 A function F defined on $(-a, a)$ is said to be odd if

$$F(x) = -F(-x) \quad \text{for all } x \in (-a, a).$$

(i) Let f be an odd function defined on $(-a, a)$ such that its inverse exists and the Maclaurin series for $f(x)$ exists.

(a) By writing down the Maclaurin series for $f(x)$ and $f(-x)$ or otherwise, show that the Maclaurin series for $f(x)$ contains only odd powers of x . [3]

(b) Show that its inverse function f^{-1} is also an odd function. [2]

(ii) Let $g(x) = 2 \tan x - x$, $|x| < \frac{\pi}{2}$.

(a) Show that g is an odd function and its inverse exists. [2]

(b) Find the Maclaurin series for $g(x)$ up to the term in x^5 . [3]

(c) Hence find the Maclaurin series for $g^{-1}(x)$ up to the term in x^5 . [3]

2 Let P_n denote the set of prime numbers less than or equal to n , $n \geq 2$.

(i) Explain why for each $p \in P_n$,

$$\frac{1}{1 - \frac{1}{p}} > 1 + \frac{1}{p} + \frac{1}{p^2} + \dots + \frac{1}{p^n}. \quad [1]$$

(ii) Hence deduce that

$$\prod_{p \in P_n} \frac{1}{1 - \frac{1}{p}} > \sum_{j=1}^n \frac{1}{j}. \quad [2]$$

(iii) Justify the following statements:

(a) $\sum_{j=1}^n \frac{1}{j} > \ln n$ for all positive integers n , [3]

(b) $\frac{7}{5}x > \ln \frac{1}{1-x}$ on $\left(0, \frac{1}{2}\right]$. [3]

(iv) Hence show that $\sum_{p \in P_n} \frac{1}{p} > \frac{5}{7} \ln(\ln n)$ and deduce that there are infinitely many prime numbers. [3]

- 3 (i) Show that the number of ways to select 5 non-empty subsets $A_1, A_2, \dots, A_5$ of $\{1, 2, \dots, 9\}$ such that
- $|A_i \cap A_j| = 0$, for all $i \neq j$, and
 - $A_1 \cup A_2 \cup \dots \cup A_5 = \{1, 2, \dots, 9\}$

is given by $5^9 - \binom{5}{1}4^9 + \binom{5}{2}3^9 - \binom{5}{3}2^9 + \binom{5}{4}1^9$. [3]

Let $S(r, n)$ denote the number of ways to distribute r distinct objects into n identical boxes such that no box is empty.

- (ii) Using (i) or otherwise, find $S(9, 5)$. [2]

- (iii) Explain why for $r \geq 3$, $S(r, 3) = \frac{1}{2}(3^{r-1} + 1) - 2^{r-1}$. [3]

- (iv) Determine the number of ways to select two non-empty subsets of $\{1, 2, \dots, n\}$ such that one is not a subset of the other, and the intersection of these two sets is the null set. [3]

- 4 (i) It is given that A , B and C are real numbers. Show that the quadratic function $h(t) = At^2 + 2Bt + C$ is non-negative for all real values of t if and only if $A > 0$ and $AC \geq B^2$. [2]

- (ii) Let f and g be continuous functions on the interval $[a, b]$. By considering $\int_a^b (tf(x) + g(x))^2 dx$, show that

$$\left(\int_a^b (f(x))^2 dx \right) \left(\int_a^b (g(x))^2 dx \right) \geq \left(\int_a^b f(x)g(x) dx \right)^2$$

and determine when equality holds. [3]

- (iii) Hence show that $\int_0^1 \sqrt{1+x^3} dx < \frac{\sqrt{5}}{2}$. [3]

- (iv) Let k be a continuous and differentiable function on the interval $[0, 1]$ satisfying $k(1) = 0$. Show that

$$\left(\int_0^1 k(x) dx \right)^2 \leq \frac{1}{3} \left(\int_0^1 (k'(x))^2 dx \right)$$

and determine when equality holds. [4]

- 5 For any positive integer n , the function $f_n(x)$ is defined by

$$f_n(x) = \frac{x(x+1)(x+2)\dots(x+n-1)}{n!}.$$

- (a) By considering $f_n(x+1) - f_n(x)$ or otherwise, show that $\sum_{n=1}^N f_n(x) = f_N(x+1) - 1$. [3]

- (b) Show that for all positive integers N , $0 < f_N\left(\frac{1}{2}\right) < \frac{1}{\sqrt{2N+1}}$. [5]

Hence or otherwise, show that the infinite series $\frac{1}{2 \times 4} + \frac{1 \times 3}{2 \times 4 \times 6} + \frac{1 \times 3 \times 5}{2 \times 4 \times 6 \times 8} + \dots$ is convergent and find its sum. [3]

- 6 A lattice point is a point such that its coordinates are all integers.

- (i) Prove that among any set of five lattice points $(x_1, y_1), (x_2, y_2), \dots, (x_5, y_5)$ in the plane, there must be two of them whose midpoint is also a lattice point. [3]

- (ii) Determine with proof, the least m such that among any set of m lattice points $(x_1, y_1, z_1), (x_2, y_2, z_2), \dots, (x_m, y_m, z_m)$ in space, there must be two of them whose midpoint is also a lattice point. [3]

- (iii) The centroid of 4 lattice points $(x_a, y_a), (x_b, y_b), (x_c, y_c), (x_d, y_d)$ is defined by

$$\left(\frac{x_a + x_b + x_c + x_d}{4}, \frac{y_a + y_b + y_c + y_d}{4} \right).$$

Using (i) or otherwise, prove that among any set of 13 lattice points $(x_1, y_1), (x_2, y_2), \dots, (x_{13}, y_{13})$ in the plane, there must be four of them whose centroid is also a lattice point. [3]

A lattice point (x, y) is (a, b) modulo n if and only if

$$x \equiv a \pmod{n} \quad \text{and} \quad y \equiv b \pmod{n}.$$

- (iv) Let S be a set of twelve lattice points in the plane such that three of its points are $(0, 0)$ modulo 4, three are $(0, 1)$ modulo 4, three are $(1, 0)$ modulo 4 and three are $(1, 1)$ modulo 4. Show that there does not exist 4 points of S whose centroid is a lattice point. [2]

- 7 A string of $2n$ parenthesis with n open '(' and n closed ')' parentheses is said to be valid if each open parenthesis has a matching closed parenthesis occurring after it. Otherwise, it is invalid. For example, when $n = 2$, $(())$ and $()()$ are the only two valid strings while $())($ is an invalid string.

Let C_n be the number of valid strings of n matching pairs of parentheses and define $C_0 = 1$.

- (a) (i) Find C_1 and C_3 . [2]

- (ii) Use the bijection principle to show that the number of invalid strings with n open and n closed parentheses is the same as the number of strings with $n-1$ open and $n+1$ closed parentheses and find this number in terms of n . [3]

- (iii) Hence or otherwise, show that $C_n = \frac{1}{n+1} \binom{2n}{n}$. [2]

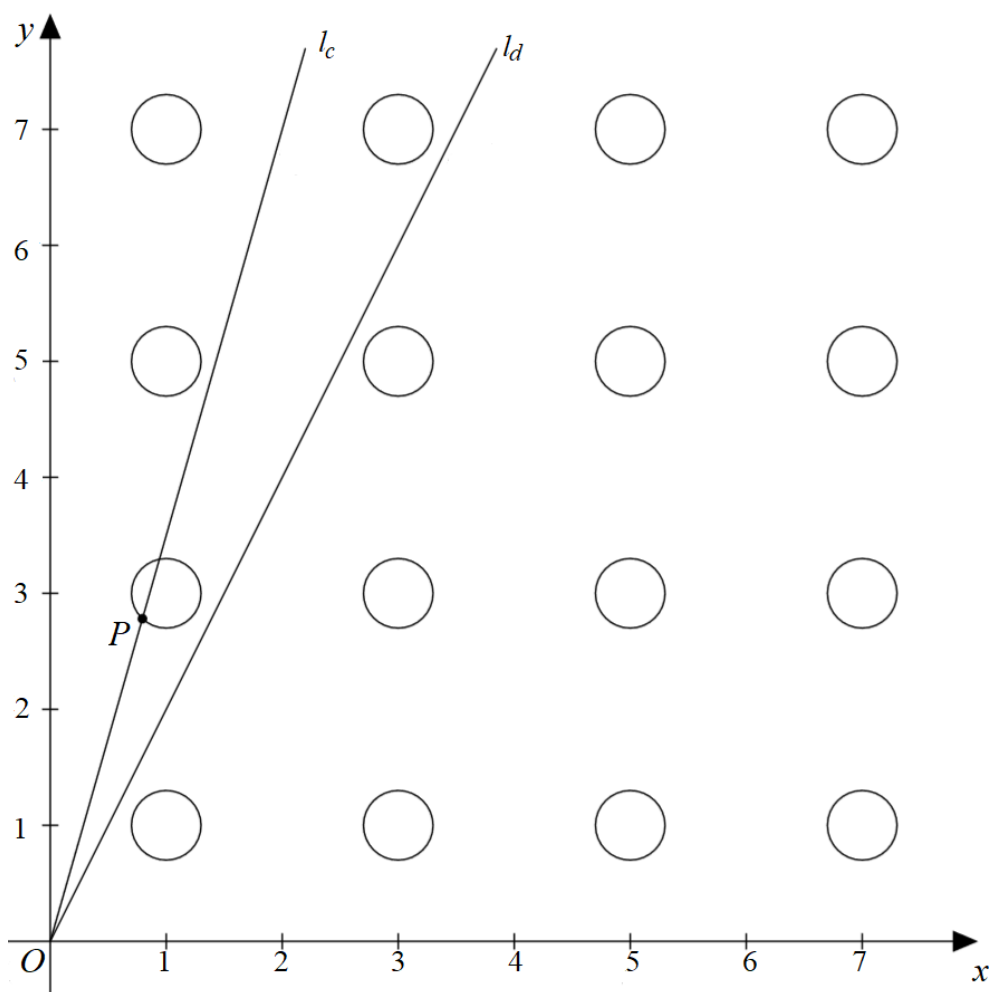
- (b) (i) For any positive integer n , explain why

$$C_n = C_{n-1}C_0 + C_{n-2}C_1 + \cdots + C_1C_{n-2} + C_0C_{n-1}. \quad [3]$$

- (ii) Hence show that if C_n is odd, n and $\frac{C_{n-1}}{2}$ are both odd. [3]

- (iii) Hence determine all positive integers n such that C_n is odd. [3]

- 8 Taylor is at the origin O , lost in the middle of the woods in which the trunks of the trees all have the same radius $R > 0$ and are represented by circles centered at the coordinates (a, b) , where a and b are odd integers.



To get out of the woods, Taylor travels along l_m , the half-line defined by $y = mx$ for $x \geq 0$. For example, in the figure above, l_c meets the tree centered at $(1, 3)$ at P , and Taylor is unable to move past P . However, Taylor is able to move past the first 2 columns of trees if she moves along l_d .

- (i) Let m be a positive real number. Show that l_m meets the tree with radius $R > 0$ centered at (a, b) if and only if $|b - ma| \leq R\sqrt{1 + m^2}$. [3]

It is known that any irrational number m can be well approximated by a rational number $\frac{b}{a}$ whose numerator and denominator are both odd integers. In other words, for any $\varepsilon > 0$, there exists two positive odd numbers a and b such that $|b - ma| \leq \varepsilon$.

- (ii) Hence explain why if m is irrational, then l_m meets a tree. [1]

Question 8 continues on the next page

- (iii) Suppose that $m = \frac{p}{q}$, where p and q are positive coprime integers.
- (a) Suppose that p and q are odd. Show that l_m meets a tree, explaining clearly your reasoning. [1]
- (b) Suppose that p and q are of different parity and that l_m meets a tree. Show that $1 \leq R\sqrt{p^2 + q^2}$. [2]
- (iv) Suppose $R \geq \frac{1}{\sqrt{5}}$. Show, in any order, that the half-lines l_m meets the tree centered at
- (a) $(1, 1)$ if $\frac{1}{2} \leq m \leq 2$,
- (b) $(1, \alpha)$, where α is the closest odd integer to m if $m > 2$,
- (c) $(\alpha, 1)$, where α is the closest odd integer to m if $m < \frac{1}{2}$. [5]

We say that Taylor can *get out of the woods* if there exists a half-line l_m which does not meet any of the circles.

- (v) Show that if Taylor can get past the first row or first column of trees, Taylor can *get out of the woods*. [2]

[END OF PAPER]